



COMILLAS
UNIVERSIDAD PONTIFICIA

ICAI

ICADE

CIHS

ICADE International

Fall 2026

Financial Markets

L3 – Exercises and Solutions
September 30th, 2026

comillas.edu

Maturity, price and duration

Calculate duration of following cash flow using a discount of 15%:

Tiempo	1	2	3
Flujo de caja	3.000,00	2.000,00	1.000,00

$$duration = \frac{1 \cdot \frac{3.000}{1,15} + 2 \cdot \frac{2.000}{1,15^2} + 3 \cdot \frac{1.000}{1,15^3}}{\frac{3.000}{1,15} + \frac{2.000}{1,15^2} + \frac{1.000}{1,15^3}} = 1,59 \text{ years}$$

Calculate duration of a **zero coupon bond** with maturity of 10 years and a **yield to maturity** of 5%. Also calculate the issue price and the **discount**. (Nominal=1.000€)

$$duration = maturity = 10 \text{ years}$$

$$Price = \frac{1.000}{1,05^{10}} = 613,91$$

$$Discount = \frac{1.000 - 613,91}{1.000} = 38,61\%$$



Maturity, price and duration

A Bond is issue without discount, with maturity of 3 years, facial of 1.000€ and three coupons of 100€ to be paid yearly. Calculate yield to maturity and duration (with IRR)

Time		1	2	3
Debt	- 1.000,00			1.000,00
Coupon		100,00	100,00	100,00
Cash Flow	- 1.000,00	100,00	100,00	1.100,00

$$duration = \frac{1 \cdot \frac{100}{1,1} + 2 \cdot \frac{100}{1,1^2} + 3 \cdot \frac{1.100}{1,1^3}}{\frac{100}{1,1} + \frac{100}{1,1^2} + \frac{1.100}{1,1^3}} = 2,74 \text{ years}$$

$$IRR \Rightarrow -1.000 + \frac{100}{1+r} + \frac{100}{(1+r)^2} + \frac{1.100}{(1+r)^3} = 0 \Rightarrow IRR = 10\% \Rightarrow \frac{100}{1,1} + \frac{100}{1,1^2} + \frac{1.100}{1,1^3} = 1.000$$

A Bond is issue without discount, with maturity of 10 years, facial of 1.000€ and ten coupons of 100€ to be paid yearly. Calculate yield to maturity and duration (with IRR)

Time		1	2	3	4	5	6	7	8	9	10
Debt	- 1.000,00										1.000,00
Coupon		100,00	100,00	100,00	100,00	100,00	100,00	100,00	100,00	100,00	100,00
Cash flow	- 1.000,00	100,00	100,00	100,00	100,00	100,00	100,00	100,00	100,00	100,00	1.100,00

	Total										
NPV	1.000,00	90,91	82,64	75,13	68,30	62,09	56,45	51,32	46,65	42,41	424,10
time x NPV	6.759,02	90,91	165,29	225,39	273,21	310,46	338,68	359,21	373,21	381,69	4.240,98
Duration	6,76										



Interest rate risk

A government Bond is issued without discount, with maturity of 10 years, face of 1.000€ and ten coupons of 20€ to be paid yearly. The interest rate is 2%

$$IRR \Rightarrow -1.000 + \frac{20}{1+r} + \frac{20}{(1+r)^2} + \dots + \frac{1.020}{(1+r)^{10}} = 0 \Rightarrow IRR = 2\%$$

If the bond is sold after one year time and the interest rate remains unchanged the return is:

$$Value = \frac{20}{1,02} + \frac{20}{1,02^2} + \dots + \frac{1.020}{1,02^9} = 1.000$$

$$\text{return} = \frac{1.000 + 20 - 1.000}{1.000} = 2\%$$

i=2%

...but... if interest rate rises to 3% or falls to 1%

$$Value = \frac{20}{1,03} + \frac{20}{1,03^2} + \dots + \frac{1.020}{1,03^9} = 922,14$$

$$\text{return} = \frac{922,14 + 20 - 1.000}{1.000} = -5,79\%$$

i=3%

$$Value = \frac{20}{1,01} + \frac{20}{1,01^2} + \dots + \frac{1.020}{1,01^9} = 1.085,66$$

$$\text{return} = \frac{1.085,66 + 20 - 1.000}{1.000} = 10,57\%$$

i=1%

Coupons Twice a Year — Example

Almost every real bond pays twice a year, not once

A bond has face value \$1,000, a coupon rate of 8% paid semi-annually, three years to maturity, and is quoted at a yield of 6%

What do you discount, at what rate, and over how many periods?

What is the price of the bond?

Coupons Twice a Year — Answer

Per period: coupon \$40, yield 3%, six periods. Halve both, double the count

$$P = 40 \times [1 - 1.03^{-6}] / 0.03 + 1,000 / 1.03^6$$

$$P = 216.69 + 837.48 = \$1,054.17$$

Coupon rate above market yield, so the bond trades at a premium. As always

Nothing new: the same discounting, with the clock running twice as fast

The Quoted Rate Is Not the Rate You Earn — Example

That bond was quoted at 6%. It pays 3% every six months

If you reinvest the first half-year coupon, what do you actually earn in a year?

A credit card quotes an APR of 18%, charged monthly. What do you pay in a year?

APR is a convention: the periodic rate multiplied by the number of periods

EAY is the truth: what a year actually costs once interest earns interest

The Quoted Rate Is Not the Rate You Earn — Answer

Bond: $EAY = (1.03)^2 - 1 = 6.09\%$, not 6%

Card: $EAY = (1.015)^{12} - 1 = 19.56\%$, not 18%

$EAY = (1 + APR/m)^m - 1$, where m is the number of periods per year

The more often interest is charged, the wider the gap

Compare two investments by their EAY, never by their quoted rate

Term Structure of Interest Rates (TSIR).

Zero Coupon curve (II)

A bond is equivalent to a surface of zero coupon bonds with same payments.

$$IRR_{1\text{ year}} = -0,27\% \rightarrow ZCC_{1\text{ year}} = -0,27\%$$

$$IRR_{2\text{ year}} = -0,13\%$$

$$IRR_{3\text{ year}} = 0,00\%$$

$$1.000 = \frac{-1,3}{0,9973} + \frac{998,68}{(1+ZCC_2)^2} \rightarrow ZCC_2 = -0,13\%$$

$$1.000 = \frac{0}{0,9973} + \frac{0}{(1+ZCC_2)^2} + \frac{1.000}{(1+ZCC_3)^3} \rightarrow ZCC_3 = 0,00\%$$

$$IRR_3 = 5\%$$

$$1.000 = \frac{50}{0,9973} + \frac{50}{(1+ZCC_2)^2} + \frac{1.050}{(1+ZCC_3)^3} \rightarrow ZCC_3$$

Year	Zero CC	IRR
1	-0,27%	-0,27%
2	-0,13%	-0,13%
3	0,00%	0,00%
4	0,13%	0,13%
5	0,25%	0,25%
6	0,37%	0,36%
7	0,48%	0,48%
8	0,59%	0,58%
9	0,70%	0,69%
10	0,80%	0,78%



The Expectations Hypothesis — Example

The one-year yield is 2% and the two-year yield is 3%

What does the market expect the one-year rate to be next year?

Assume investors are risk neutral

Now suppose investors come to expect 5% instead

What happens to today's two-year yield, and to the bond you already own?

The Expectations Hypothesis — Answer

$$f = (1.03)^2 / 1.02 - 1 = 4.01\%$$

roughly $2 \times 3 - 2$: the long rate is an average, so year two carries the difference

If they expect 5%: $(1 + y_2)^2 = 1.02 \times 1.05 = 1.071$, so $y_2 = 3.49\%$

The two-year yield rises, so the two-year bond you own falls in price

Nobody defaulted and no central bank met. An expectation changed. That was enough

The Growing Perpetuity — Check

A flat is rented for \$12,000 next year

Rents grow 2% a year for ever. You require 6%

$$PV = 12,000 / (0.06 - 0.02) = \$300,000$$

If rents grew 3% instead: $12,000 / 0.03 = \$400,000$

One point of growth, a third more value

Remember that number. It comes back with Gordon

Fundamental vs Actual Value — Example

A share trades today at \$48

Next year's dividend is expected to be \$4, and next year's price is expected to be \$52

The return required, given its risk, is 8%

- (a) What is the share's fundamental value?
- (b) What return do you expect to earn?
- (c) Is it over- or undervalued?

Fundamental vs Actual Value — Answer

(a) $V_0 = (4 + 52) / 1.08 = \51.85

(b) $E[HPR] = 56 / 48 - 1 = 16.67\%$

(c) It trades at 48 and is worth 51.85: undervalued

Or: you expect 16.67% where 8% would do

Those two statements are the same statement

A share is cheap when its expected return beats
the return it is required to earn

And notice: nobody told us where the 8% came from

Zero Growth — Example

JPMorgan Chase has a perpetual preferred share that pays a dividend of \$1.20 a year, for ever

(a) What is it worth if the required return is 6%?

(b) The share actually trades at \$15.

What return is the market expecting?

Zero Growth — Answer

(a) $V_0 = D / r = 1.20 / 0.06 = \20.00

a fixed dividend with no maturity is a perpetuity

(b) $r = D / P_0 = 1.20 / 15 = 8.00\%$

- Same instrument, same \$1.20. Pay less and you earn more
- Exactly the relationship you know from bonds
- Which number is the truth, \$20 or \$15?
- That is market efficiency, and it comes later



Gordon — Example

A share is expected to pay a dividend of \$2 next year

Its long-term growth rate is 4% and the required return is 8%

(a) What is its fundamental value?

(b) What happens if growth slows to 3%?

(c) And if instead the required return rises to 9%?

Gordon — Answer

(a) $V_0 = 2 / (0.08 - 0.04) = \50.00

(b) $2 / (0.08 - 0.03) = \$40.00$, a fall of 20%

(c) $2 / (0.09 - 0.04) = \$40.00$, again a fall of 20%

One point, from either side, costs a fifth of the firm

The denominator is small, so small changes in it
are large changes in the price

This is why share prices move so violently on news
that changes nothing at all this year



Where Growth Comes From — Example

A firm will earn \$5 a share next year and pay out 40% as dividends

It reinvests the rest at a return on equity (ROE) of 15%. Investors require 12%

- (a) What is the growth rate g ?
- (b) What is the share worth?
- (c) What would it be worth if it paid out everything and never grew?
- (d) And if ROE were 12% instead?

Where Growth Comes From — Answer

- (a) $g = \text{plowback} \times \text{ROE} = 0.60 \times 15\% = 9\%$
- (b) $D1 = 0.40 \times 5 = \$2$. $V0 = 2 / (0.12 - 0.09) = \66.67
- (c) No growth: $V0 = E1 / r = 5 / 0.12 = \41.67
- $PVGO = 66.67 - 41.67 = \$25.00$: what growth is worth

- (d) $g = 0.60 \times 12\% = 7.2\%$. $V0 = 2 / 0.048 = \$41.67$. $PVGO = 0$
- Growth creates value only when ROE is above r

